THERMAL PROPERTIES OF MATTER

FORMULA SHEET | JEE MAINS | Let's Play With Physics

1. TEMPERATURE SCALES

$$\frac{C-0}{100}=\frac{F-32}{180}=\frac{K-273.15}{100}=\frac{X-\text{LFP}}{\text{UFP}-\text{LFP}}$$

$\Delta T_C=\Delta T_K,\quad \Delta T_F=\frac{9}{5}\Delta T_C$
$-40^\circ\text{C}=-40^\circ\text{F};\quad 574.59\text{ K}=574.59^\circ\text{F}$

⋆ Common Pitfall

- A temperature **difference** of 1°C equals 1 K, but 1°C ≠ 1 K as a reading
- Use the general formula for any arbitrary scale (LFP = ice point, UFP = steam point)

2. THERMAL EXPANSION

$$\Delta L=L\alpha\Delta T,\quad \Delta A=A\beta\Delta T,\quad \Delta V=V\gamma\Delta T$$
$$\alpha:\beta:\gamma=1:2:3\qquad(\beta=2\alpha,\ \gamma=3\alpha)$$

Density change: $\rho'=\dfrac{\rho}{1+\gamma\Delta T}\approx\rho(1-\gamma\Delta T)$

Liquid in vessel: $\gamma_{\text{real}}=\gamma_{\text{app}}+\gamma_{\text{vessel}}$

Ideal gas at constant P: $\gamma=\dfrac{1}{T}$

Thermal stress (rod clamped): $\dfrac{F}{A}=Y\alpha\Delta T$

Pendulum clock: $\dfrac{\Delta T}{T}=\dfrac{1}{2}\alpha\Delta\theta$
Time lost/gained per day $=\frac{1}{2}\alpha\Delta\theta\times 86400$ s

Bimetallic strip: bends with higher-α metal on the convex side when heated

Constant length difference: $L_1\alpha_1=L_2\alpha_2$

Anomalous expansion of water: density is max at 4°C; water contracts from 0°C to 4°C

⋆ Common Pitfall

- A hole in a plate **expands** on heating, as if made of the same material
- Clock gains time in winter, loses in summer
- Volume expansivity of an ideal gas at constant P is $\gamma=1/T$ (decreases with T)

3. CALORIMETRY

$$Q=mc\Delta T\qquad Q=mL\qquad C=mc\text{ (heat capacity)}$$

Principle of calorimetry: heat lost = heat gained

Quantity	Value
c_{water}	1 cal/g°C = 4186 J/kg K
c_{ice}	0.5 cal/g°C = 2100 J/kg K
L_f (ice)	80 cal/g = 3.36×10^5 J/kg
L_v (water)	540 cal/g = 2.26×10^6 J/kg
1 cal	4.186 J

Water equivalent: $W=\dfrac{mc}{c_{\text{water}}}$ (in grams of water)

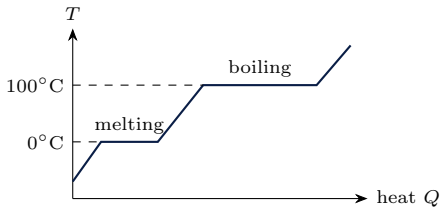
Molar specific heat: $C=Mc$; solids (Dulong–Petit): $C\approx 3R$

Ice + water mixing: first check heat needed to melt all ice (m_iL_f) vs heat available from water cooling to 0°C. If insufficient, final $T=0^\circ$C with some ice left.

⋆ Common Pitfall

- During phase change temperature stays constant
- Never assume all ice melts — always compare heats first
- Latent heat of vaporisation > fusion; steam burns are worse than boiling water burns

4. HEATING CURVE SecHEATING CURVE & PHASE DIAGRAM CHANGE OF STATE



Slope of T–$Q=\dfrac{1}{mc}$; length of flat part $=mL$

Triple point of water: 273.16 K, 0.006 atm

⋆ Common Pitfall

- Steeper slope ⇒ smaller specific heat
- Boiling point increases with pressure (pressure cooker); melting point of ice decreases with pressure (regelation)

5. CONDUCTION

$$H=\frac{dQ}{dt}=\frac{kA\,\Delta T}{L},\qquad R_{\text{th}}=\frac{L}{kA}$$
$$H=\frac{\Delta T}{R_{\text{th}}}\quad(\text{like } I=V/R)$$

Series: $R=R_1+R_2,\ k_{\text{eq}}=\dfrac{L_1+L_2}{\frac{L_1}{k_1}+\frac{L_2}{k_2}}$ (equal lengths: $\dfrac{2k_1k_2}{k_1+k_2}$)

Parallel: $\dfrac{1}{R}=\dfrac{1}{R_1}+\dfrac{1}{R_2},\ k_{\text{eq}}=\dfrac{k_1A_1+k_2A_2}{A_1+A_2}$

Junction temperature (series): $\theta=\dfrac{\frac{k_1}{L_1}\theta_1+\frac{k_2}{L_2}\theta_2}{\frac{k_1}{L_1}+\frac{k_2}{L_2}}$

Ice layer growth: $t=\dfrac{\rho L}{2k\theta}\left(x_2^2-x_1^2\right)$

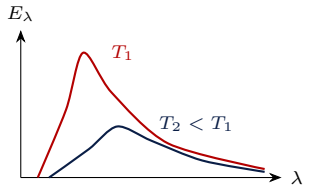
⋆ Common Pitfall

- In steady state H is the same through every layer in series
- Time for ice thickness to go $0\to x$ and $x\to 2x$ is in ratio 1 : 3

6. RADIATION

Stefan–Boltzmann: $P = e\sigma AT^4$
Net loss: $P = e\sigma A\left(T^4 - T_0^4\right)$
$\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}
Wien's law: $\lambda_m T = b = 2.898 \times 10^{-3}$ m K

Kirchhoff's law: good absorber = good emitter; $\dfrac{E}{a}$ is the same for all bodies at a given T (equals $E_{\text{black body}}$)



Area under curve $\propto T^4$; peak shifts left as T rises

⋆ Common Pitfall

- $\lambda_m \propto 1/T$; $P \propto T^4$ — doubling T gives 16× power
- Radiation needs no medium; fastest mode of heat transfer

7. NEWTON'S LAW OF COOLING

$$\frac{dT}{dt} = -k(T - T_0) \;\Rightarrow\; T - T_0 = (T_i - T_0)e^{-kt}$$

Average form:

$$\frac{T_1 - T_2}{t} = k\left[\frac{T_1 + T_2}{2} - T_0\right]$$

$k = \dfrac{4e\sigma AT_0^3}{mc}$ (valid for small temperature difference)

⋆ Common Pitfall

- Rate of cooling depends on surface area & nature, not on volume alone; same material spheres: rate $\propto 1/r$
- Newton's law holds only for small $(T - T_0)$; otherwise use Stefan's law

8. JEE SHORTCUTS & MASTER TIPS

1. $\beta = 2\alpha$, $\gamma = 3\alpha$
2. Clock error per day $= \frac{1}{2}\alpha\Delta\theta \times 86400$ s
3. Thermal stress $= Y\alpha\Delta T$ (independent of length)
4. Conduction: treat like resistors, $R = L/kA$
5. Wien: $\lambda_m T = 2.9 \times 10^{-3}$ m K
6. Newton's cooling: use average form for MCQs with two intervals

⋆ Master Confusion List

1. **Series vs parallel** thermal rods: series → add R
2. **Specific heat vs heat capacity:** c per kg; $C = mc$ for the body
3. **Ice mixing:** check before assuming full melting
4. **Wien vs Stefan:** $\lambda_m \propto 1/T$ vs $P \propto T^4$
5. **Hole expands** on heating
6. **Water** is densest at 4°C, not 0°C

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